Abraar Ahmad

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Intro

- Aspiring AI/ML student with hands-on experience building machine learning and deep learning projects. Focused on solving real-world problems through practical model development and experimentation. Seeking an opportunity to contribute to impactful AI/ML-driven products while continuing to strengthen my technical foundation.

Education

Inderprastha Engineering College — B.Tech Computer Science Oct 2021 – May 2025
Affiliated to Dr. A. P. J. Abdul Kalam Technical University

Selected Projects

ML Projects

- **Heart Disease Prediction — FastAPI + React + ML** Trained logistic regression and tree-based models, engineered features, produced reproducible notebooks and model artifact; achieved **90% CV accuracy**. (Code, notebooks, model artifact).
- **Crop Recommendation System (ML)** Built XGBoost / tree models for crop suitability; focused on feature engineering, handling imbalanced data and model interpretability.
- **Breast Cancer Prediction — KNN / XGBoost** End-to-end pipeline for diagnosis prediction with evaluation notebooks, cross-validation and baseline comparisons.

What I Learned (Machine Learning Foundations) — Through these ML projects, I strengthened my understanding of supervised learning workflows — from data preprocessing, feature engineering, and handling missing values to model selection, cross-validation, hyperparameter tuning, and performance evaluation.

I learned how different algorithms behave (Logistic Regression, KNN, XGBoost), how the bias–variance tradeoff impacts generalization, and why proper validation strategies are critical. I gained practical experience in fine-tuning models using techniques such as grid search and cross-validation to improve performance systematically rather than relying on default parameters.

I also developed intuition for interpreting metrics like precision, recall, F1-score and confusion matrices depending on the problem context (e.g., healthcare vs. recommendation systems).

Deep Learning Projects

- **Building n-layers Neural Network from Scratch** — implemented forward/backward passes, activation functions and training loop from first principles.
- **ANN classification — end-to-end** — complete pipeline: preprocessing, network design, training, evaluation and visualization.
- **Simple Q&A using RNN in PyTorch** — RNN approach for sequence modeling and toy Q&A tasks.
- **Next-word predictor using LSTM (PyTorch)** — LSTM-based language model for next-word prediction, focused on sequence preprocessing and sampling.
- **Transformer encoder–decoder from scratch (PyTorch)** — custom implementation of attention mechanisms and encoder-decoder flow.
- **NanoGPT from scratch (PyTorch)** — training small transformer/GPT style model

What I learned (deep learning / transformers) — I learned transformers from a research perspective using first principles. I started by deeply understanding RNNs — why they were designed, how they work, and their limitations — then studied LSTMs (their motivations and mechanics). From there I worked through sequence-to-sequence learning (Sutskever et al.), then attention and alignment (Bahdanau et al.), and the Luong et al. improvements. Finally I studied "Attention Is All You Need" intensively until my intuition became strong enough that when I use GPT-style models I can visualise what's happening under the hood. I have now built a NanoGPT-based model and plan to learn and apply transfer-learning techniques (as suggested by Jeremy Howard)

to improve performance. (i've more projects you can check them on my gitHub: github.com/Abraar77)

Technical Skills

Languages / Core: Python, NumPy, Pandas
NLP / Research: NLP fundamentals, sequence modeling, attention mechanisms
ML & DL: PyTorch, scikit-learn, XGBoost, Transformers
Tools: Jupyter, Git

Availability

Why Me

am genuinely interested in research and want to make a societal impact. Recently I considered switching to more product-oriented/agentic AI work because I thought research roles are hard to break into at the junior level — then I saw your job posting on LinkedIn and it felt like someone reignited my purpose. I am motivated to contribute, learn rapidly, and take on research experiments that move projects forward.I am adaptable, quick to learn, and comfortable working independently on challenging problems. I am eager to contribute value while continuously improving under mentorship and real-world exposure.

Strengths

- **Critical Thinking** — I naturally question assumptions and focus on understanding the **why**, **what**, and **how** behind models and architectures rather than just implementing them. This allows me to debug systematically and improve model performance logically. I believe this mindset stems from my long-standing interest in philosophy, which trained me to analyze problems from first principles.
- **Go-Getter Mindset** — I take initiative to explore concepts independently, whether it's implementing neural networks from scratch or studying research papers deeply to strengthen my intuition.
- **Self-Disciplined** — I maintain consistent study and practice routines, balancing long-term goals with structured daily effort. This discipline reflects in my ability to complete complex projects end-to-end.